

ULTRASONIC ANEMOMETER

The Ultrasonic Anemometer is a high-performance, solid-state wind sensor designed for precision measurement of wind speed and direction without any mechanical moving parts.

Utilizing advanced ultrasonic Time-of-Flight (TOF) technology, it ensures reliable, maintenance-free operation in a wide range of environmental conditions.

Housed in a rugged, corrosion-resistant polycarbonate enclosure, this compact and lightweight sensor is ideal for IoT integration, weather monitoring, smart agriculture, UAVs, and industrial applications.



Key Features

- Non-Contact Measurement:** No moving parts ensure long-term reliability and reduced maintenance.
- High Accuracy & Resolution:** Delivers precise wind data with minimal error margins.
- Wide Operating Range:** Suitable for both indoor and outdoor environments.
- Real-Time Data Output:** Compatible with IoT platforms for live monitoring and analytics.
- Robust Construction:** IP66-rated housing resistant to dust, water, and UV exposure.
- Low Power Consumption:** Optimized for continuous operation in remote or solar-powered setups.

Technical Specifications

Wind Speed	
Parameter	Value
Range	0 – 65 m/s
Accuracy	±1%
Resolution	0.1 m/s
Response Time	Instantaneous

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Wind Direction	
Parameter	Value
Range	0° – 359°
Resolution	1°
Response Time	Instantaneous
Accuracy	±2°

Electrical Specifications	
Parameter	Value
Supply Voltage	2V – 16V DC
Power Consumption	20mW
Output Type	Digital & Analog

Pinout / Wiring	
Wire Colour	Function
Red	VCC (2V-16V DC)
Black	GND
Blue	A (RS232) / RX (RS485)
Yellow	B (RS232) / TX (RS485)

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Environmental Ratings

Parameter	Value
Ingress Protection	IP66
Operating Temperature	-40°C to +70°C

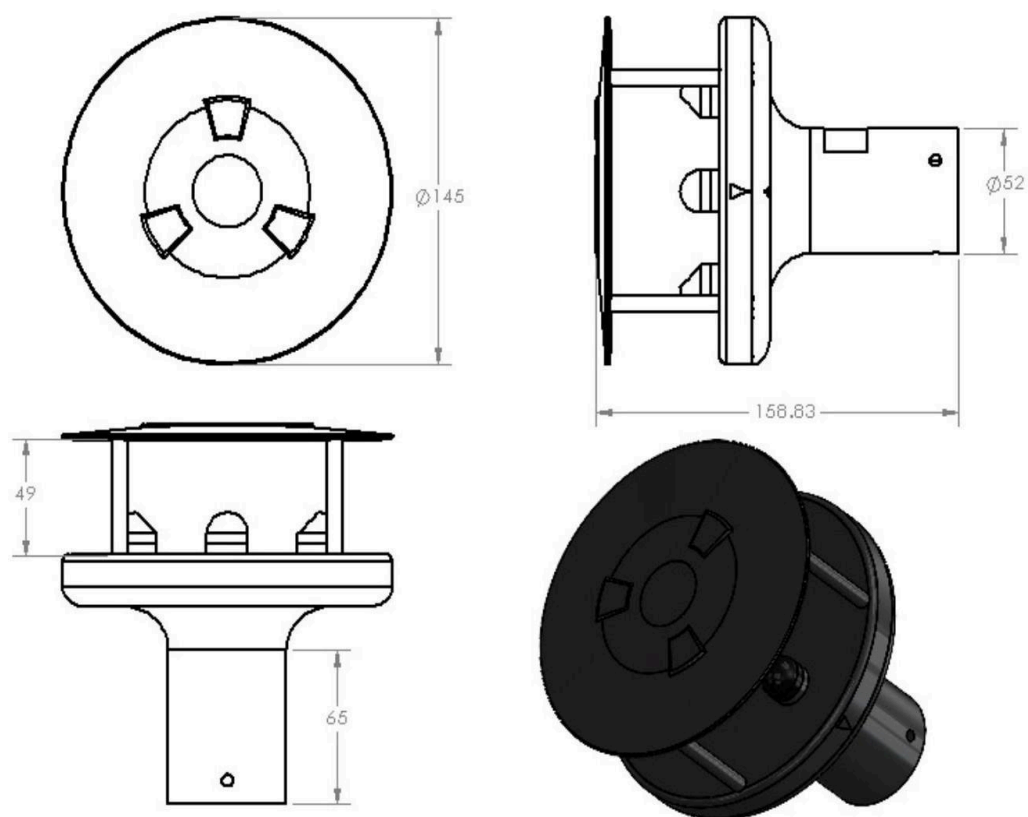
Applications

- Meteorological and weather monitoring stations
- Smart farming and precision agriculture
- IoT-based environmental sensing networks
- UAVs and drones for in-flight wind analysis
- Industrial ventilation and airflow systems
- Small airports, helipads, and wind energy projects

Mechanical Specifications

Property	Value
Housing Material	ABS (Acrylonitrile Butadiene Styrene)
Protection Rating	IP66 (dust and water resistant)
Dimensions	159mm (H) × 145mm (Ø)
Weight	340 g
Colour	Black
Mounting	Pole mount with stainless-steel mounting
Cable Connection	4-pin waterproof connector
UV Resistance	> 5 years outdoor exposure

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